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Cell- & Tissue-based Therapy

Regulatory T cells and immunity to pathogens

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Immune responses to pathogens are modulated by one or more types of cells that perform a regulatory function. Some cells with this function, such as $CD4^+Foxp3^+$ natural regulatory T cells (nTreg), pre-exist prior to infections whereas others may be induced as a consequence of infection (adaptive Treg). With pathogens that have a complex pathogenesis, multiple types of regulatory cells could influence the outcome. One major property of Treg is to help minimize collateral tissue damage that can occur during immune reactions to a chronic infection. The consequence is less damage to the host but in such situations the pathogen is likely to establish persistence. In some cases, a fine balance is established between Treg responses, effector components of immunity and the pathogen. Treg responses to pathogens may also act to hamper the efficacy of immune control. This review discusses these issues as well as the likely mechanisms by which various pathogens can signal the participation of Treg during infection.

Keywords: cytokine, Foxp3, immunity, immunopathology, infection, regulatory cell

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1. Background

Most would now agree that immune reactions to self as well as to exogenous antigens are influenced by several types of regulatory cells. The outcome is an absent, changed or diminished adaptive and, in some cases innate, immune response. With self antigens, such regulatory responses represent one means by which the host avoids tissue damaging autoreactivity. Similarly, with reactions to allergens and foreign tissue transplants regulatory T-cell responses benefit the host by preventing or diminishing the magnitude of effector reactions to the foreign antigens. With tumors, regulatory counteractions work against the hosts interests because they may blunt an otherwise effective antitumor reaction. Given the wide spectrum of disease-producing properties of microbes, not surprisingly situations are diverse with regard to the relationship of regulatory cells to the outcome of infection. Moreover, the perceived role of various types of Treg can change during the course of an infection. In fact, the relationships traverse the gamut from those where regulatory cell responses appear to benefit the host to ones where the response is judged to be detrimental. In between, there are harmonious accords where regulatory cell and effector defense mechanisms appear held in balance with the pathogen. Tissue damage is minimized and the pathogen maintains persistence and can be transmitted. Such ménage a trios scenarios were observed initially with some protozoan parasitic infections. Indeed, regulatory cell responses may be critical to the establishment of concomitant immunity, a form of resistance to reinfection that typifies many parasitic infections [1].

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2. Types of regulatory cells involved

Basically, two categories of regulatory cells can influence immunity to pathogens. These are cells with regulatory activity already present at the time of pathogen invasion and others that are induced as a consequence of the infection. The best understood in the first category are the naturally occurring CD4$^+$Foxp3^{pos} T cells (nTreg) whose main function may be to prevent the occurrence of autoimmune disease [2]. Most nTreg are thought to be generated in the thymus and they mainly recognize self-antigens. However, some may also recognize epitopes expressed by pathogens [3], although it is not known if this is true for all pathogens. Other naturally occurring regulatory cells include those with the natural killer T and γδCD4$^+$ T phenotypes, but their role in the pathogenesis of infections remains poorly understood and merits further investigation.

There are several categories of inducible cells that express regulatory activity. These include cells with the same phenotype as nTreg (CD4$^+$Foxp3^{pos} IL-10 and/or TGF-β producing), Foxp3^{neg} IL-10-producing CD4$^+$ T cells (usually called Tr1 cells), CD4$^+$TGFβ-producing T cells (some of which are called T_H3 cells) as well as cells that have the CD8$^+$ phenotype [4]. All such cells form part of the adaptive immune response and can specifically recognize pathogen antigens. However, those which produce an abundance of cytokines may also exert anti-inflammatory bystander suppression at sites where they are triggered by antigens. Recently, groups working on parasitic infections have demonstrated that highly polarized T_H1 IFN-γ-producing effector cells may become, at least briefly, IL-10 producers and that these cells may be the principal source of IL-10 involved in suppressing collateral tissue damage during immunity to chronic infection [5]. The conditions that result in the conversion of T_H1 effectors into dual IFNγ and IL-10 producers remains unclear. However, continual Ag stimulation, perhaps along with IL-12 exposure, are suggested possibilities [6].

It seems likely that whereas the response to some pathogens may be influenced by a single type of regulatory cell response, in most situations more than one type may be involved. Moreover, the relative importance of different suppressor cell types could change during the course of infection and may differ between animal strains and species. In hepatitis C virus infection, for example, at least three types of regulatory cells, nTreg, Tr1 and CD8$^+$ T cells, have been suggested to modulate the immune response [7-9]. This complex topic has many unresolved issues and controversies and more studies are warranted.

3. How do pathogens signal regulatory responses?

This remains a poorly understood issue, especially with agents that cause acute infections and in circumstances where responses are influenced by nTreg. Far better understood are the potential means by which pathogens can cause adaptive T regulatory responses. Thus, it is thought that certain pathogens can create microenvironments that foster the induction of regulatory cells from nonregulatory naïve or memory cell precursors [10]. For example, a microenvironment that produces abundant IL-10 and minimal IL-12 along with repeated TCR stimulation delivered by non-mature dendritic cells (DCs) is likely to convert naïve CD4$^+$ T cells to become Foxp3$^-$ IL-10-producing Tr1 cells [11]. Pathogens can induce IL-10 because of the type of interaction they make with DCs or macrophages. Moreover, some pathogens such as Bordetella pertussis, and perhaps others, posses ligands that can induce IL-10 from DCs [12]. Other agents, such as cytomegalovirus and Epstein–Barr virus may themselves produce homologues of IL-10 [13,14]. All such circumstances can result in the induction of Tr1 IL-10-producing Tregs that are antigen specific.

Another important cytokine that may be involved in inducing adaptive Tregs is TGFβ. Thus, *in vitro* studies clearly show that naïve and memory conventional Foxp3^{neg} T cells can be converted into Foxp3^{pos} regulators that retain the antigen specificity of the precursors [15,16]. In fact in elderly humans, where the function of the thymus has declined, such conversion, perhaps mainly from memory cells, may be the major method by which all nTreg in the elderly are maintained [17]. Many pathogens may induce TGFβ and the cytokine can be present in high amounts in certain locations such as the gut, skin and eye [3,18]. Conceivably, most of the nTreg that influence responses to chronic pathogens do represent converted cells, although this was shown not to be the case with Leishmania major where antigen-specific nTreg are found at sites of viral persistence [19].

With acute infections, signaling of nTreg to influence the immune response to the pathogen could be achieved in one or more ways. First, there is the possibility that for some pathogens, and perhaps more so in some individuals, nTreg are present whose T-cell receptor repertoire recognizes antigens expressed by the agent. This subpopulation becomes the one that modulates the response. This idea has few enthusiasts at least for acute infections [20]. Some pathogens, however, do possess ligands that deliver a survival/expansion signal to nTreg. This has been suggested to be the case with HIV [21]. In addition, direct infection of nTreg, as seems to occur with feline immunodeficiency virus, results in an enhancement of the regulatory function [22]. Similarly, direct infection of DC may create a situation that favors the generation of nTreg as may occur with Friend retrovirus [23] and perhaps also with lymphocytic choriomenigitis virus [24]. It is also conceivable that a few pathogens possess epitopes that are cross-reactive with anti-self nTreg.

Other and perhaps more likely mechanisms by which pathogens signal a Treg response involve the activity of non-antigen-specific factors. Thus, the cytokines and chemokines induced in the vicinity of infection may serve to recruit, expand and activate nTreg. For example, the difference

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in susceptibility to simian immunodeficiency virus (SIV) infection of African Green monkeys (AGM) (resistant) and macaques (susceptible) seems to be in the nature of the acute phase response [25]. Resistant AGM develop an anti-inflammatory type response and the early expression of nTreg, whereas the macaques generate a proinflammatory response that delays the arrival or blunts the function of nTreg [25]. The generation of chemokines at the tissue or reactive lymphoid sites could recruit nTreg from the blood stream as the cells are known to express one or more chemokine receptors [26]. Exposure to some reactive molecules at lymphoid sites could also cause nTreg to express tissue-seeking molecules, such as the integrin αEβ7 [27].

Another potential mechanism to trigger regulatory activity is that complex pathogens could posses one or more molecules that nonspecifically activate nTreg. Candidates include Toll-like receptor (TLR) ligands, such as flagellin, shown to engage TLR5 on human nTreg and result in their activation [28]. Although nTreg may contain multiple TLRs, engagement of some of them may lead to downregulation of function as has been reported to occur with TLR2 [29]. At present, little is known about any role for other types of innate immune stimulatory ligands on the function of nTreg. Such information needs to be collected.

4. Role of regulatory T cells during pathogenesis of infections

The topic remains an evolving story and in reality only a small minority of pathogens have been studied. Published reports also tend to reflect circumstances where Treg have a measurable effect on immunity and the outcome of infection by pathogens. For most pathogens, the role of Treg over the course of infection remains unsettled and in fact is difficult to assess in humans for ethical reasons. Basically, as already mentioned, the influence of Treg can be judged either to interfere with the efficacy of immunity or to benefit the host by minimizing the collateral tissue damage that often accompanies the execution of antipathogen responses. In a few instances Treg appear to be essential participants in a balanced response that benefits both the host and the pathogen.

5. Blunting immunity

Studies of several pathogens have concluded that the presence of Treg may interfere with immunity and infection control [30]. Much of the evidence is indirect showing, for example, that antigen-driven lymphocyte proliferative or cytokine responses become enhanced if cells with the Treg phenotype are removed from the responder cell population prior to antigen stimulation [30]. Inhibitory effects on effector function of $CD8^+$ T cells have also been demonstrated and in some situations this suppression is largely antigen specific [31]. In some chronic infections, such as hepatitis C and B, the frequency of Treg in the bloodstream may be elevated compared with those that successfully control the infection. It is assumed that the Treg may be responsible for blunting the efficacy of antiviral immunity and this explains chronicity. However, this concept was not supported with experimental infections with hepatitis C virus (HCV) in chimps [32]. In these studies nTreg responses of similar magnitude were demonstrated in both recovered as well as persistently infected animals with chronic disease. Clearly further studies are warranted to evaluate these issues.

Perhaps of more relevance in chronic infections is whether or not the differential Treg activity in individuals soon after infection helps explain the eventual outcome. For example, some persons (20 – 40%) successfully resolve HCV infection probably because they make an effective $CD4^+$ T_H1 and $CD8^+$ cytokine T-cell response [33]. However, the majority suffer chronicity and have impaired T-cell responses. At present, there are no longitudinal studies following the outcome of infection in individual patients from the acute phase correlating the types and extent of their regulatory T-cell response developed over time with the eventual outcome of infection. However, one unfortunate set of circumstances did imply that the differential generation of inducible Treg responses could have detrimental consequences. Thus, in ~ 1000 women who were accidentally infected with HCV when transfused after childbirth with contaminated Ig, those that went on to develop chronic disease had more elevated IL-10-producing inducible Treg responses than those that successfully controlled infection [34].

Blunting immunity by a Treg response has also been advocated to occur in HIV infection as well as in hepatitis B virus [31,35]. For example, in the latter, the number of $Foxp3^{pos}$ cells is increased in the blood as well as in the liver in chronic severe HBV and these cells show antigen-specific suppression of effector function *in vitro* [31]. In a macaque model of AIDS the premature induction of an immunosuppressive Treg response is advocated to limit the protective antiviral immune response and contribute to viral persistence [36]. However, others have concluded that an early Treg response may explain why AGMs resist disease expression when infected with SIV as already mentioned [25].

More solid evidence that the Treg response may serve to blunt immunity to pathogens can be obtained in experimental studies in rodents. Thus, there are several ways to manipulate the levels or function of Treg and measure the effect on immunity. The most common approach is to use antibodies that primarily effect nTreg. More recently, a sophisticated inducible genetic system was developed where Treg can be completely removed in adult animals [37]. So far these systems have not been used to probe the role of Treg in infectious disease. The affect of Treg on immunity can also be studied by comparing the outcome of infection in adoptive transfer recipients given immune cells alone or along with Treg populations. These types of experiments have shown that immunity to some viruses [38], bacteria [39] and parasites [18] may be blunted when Treg are present. For example, many

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parameters of immunity to herpes simplex virus (HSV) are elevated when Treg are depleted prior to infection [38]. Such animals may clear the virus more rapidly. Immunity to HSV and some antiparasite vaccines may also be elevated in Treg depleted compared to normal animals [40,41]. Recently, it was also shown that inactivation of Treg in susceptible mice led to enhancement of several types of immune response to Theiler's virus and this also made the animals more resistant to developing demyelinating disease [42].

In experimental malaria in mice, animals depleted of Treg develop an early protective T_H1 response and infection is more rapidly resolved [43]. Similarly, in experimental plasmodium infection of human volunteers the generation of a Treg response was shown to facilitate parasite growth by suppressing the protective proinflammatory reaction induced by the parasites [18].

6. Resistance of immune exuberance

An early review on the role Treg might play during infections emphasized that the response was largely a beneficial reaction serving to minimize collateral tissue damage associated with the attempt of the body to control infection [44]. Such responses were assumed to be more pertinent with difficult to control chronic infections. Moreover, the usual price to pay in such circumstances is that the pathogen then succeeds in establishing long-term persistence. The mechanism involved in restraining tissue damage often involve the cytokines IL-10 and TGFβ. Both of these cytokines may exert a broad range anti-inflammatory effect. These include effects on the antigen-presenting cells (APC) as well as the T cells [45]. IL-10 downregulates the expression of costimulatory molecules as well as proinflammatory cytokine production by APC. It may also directly inhibit IL-12 and TNFα production by $CD4^+$ T cells. Downregulation of APC function can also be mediated by TGFβ [46]. This cytokine may also inhibit proliferation and the production of some T-cell cytokines.

Control of excessive tissue damage has been attributed to both natural and inducible Treg. Natural Treg have a propensity to enter into inflammatory sites, although at such sites some proinflammatory molecules may serve to blunt their regulatory function. Accordingly, at least *in vitro*, nTreg effectors may not function effectively in the presence of IL-6 and perhaps other proinflammatory cytokines [47]. In consequence, nTreg may be present in lesions, but their potential inhibitory effects on immune-mediated tissue damage may not be manifest [48]. This scenario is unlikely to occur with regulatory T cells that produce an abundance of cytokines such as IL-10 and TGFβ, a characteristic of adaptive Treg but perhaps also a property of $Foxp3^+$ nTreg *in vivo*. This issue merits further investigation. Clearly in several types of infections, Treg can be shown to protect against pronounced tissue damage, but the type of regulatory cell responsible for the effect is usually not well defined.

Examples of the beneficial effect of Treg have been described with several types of pathogens [30]. In HIV, for example, progression to disease appears to be the consequence of T-cell hyperactivation with this effect being inhibited when nTreg are present [49]. In herpetic stromal keratitis, a viral-induced immunoinflammatory disease, lesions are more severe when nTreg are depleted [50]. Similarly, in several bacterial and fungal infections lesions are more severe when Treg function is inhibited [39]. In one system, for example, the severity of the early inflammatory reaction was controlled by a nTreg response, which appears to target neutrophils, whereas later on the allergy-type reaction to the infecting fungus appears to be regulated by IL-10 and TGFβ-producing Tregs that targeted T_H2 immunoinflammatory cells [51]. Recently, the severity of fungal-induced granulomatous disease in humans was shown to be influenced by $Foxp3^{pos}$ Treg [52]. In several parasitic models too, the severity of lesions is influenced by the activity of IL-10-producing Treg [1]. As mentioned already, in some systems, the cells that produce anti-inflammatory IL-10 may not be $Foxp3^{pos}$ or Tr1 cells, but be normal T_H1 effectors themselves that become, at least temporarily, high producers of IL-10 [5]. These cells may be the ones mainly responsible for the suppression of severe inflammatory reactions to some parasites [5].

7. Balance

Whether or not Tregs play a beneficial or detrimental role during responses to pathogens is often difficult to judge and will vary according to circumstances. With some pathogens, the presence of Treg may form part of a harmonious situation that benefits both the host and the pathogen. This was elegantly demonstrated by Belkaid and co-workers with leishmania major infection in C57Bl/6 mice and it may be a common event with chronic infections [53]. Following resolution of skin disease caused by the leishmania parasite an entente ensues. The parasite persists indefinitely at the infection site and is transmissible. The host benefits as it remains highly immune to reinfection. The immunity is sustained by the balance between pathogen specific $Foxp3^+$ Treg and effector T cells. If the Treg are removed, the effectors can discard the pathogen but the consequence is the loss of immunity to reinfection. Even the addition of extra pathogen-specific Treg causes discord as this results in disease reactivation and parasitic dissemination to other sites [54].

A similar balance between Treg and T effectors is likely a common event with chronic pathogens and is suspected to occur with tuberculosis (TB) in humans [55]. Thus, following TB infection, rarely does sterile immunity occur and the bacilli persist almost indefinitely with no chronic consequences. Recent studies in mice have shown that a Treg response helps prevent bacilli eradication suppressing an otherwise efficient $CD4^+$ T-cell response [55]. In humans, the fine balance between TB and the host response can be broken by

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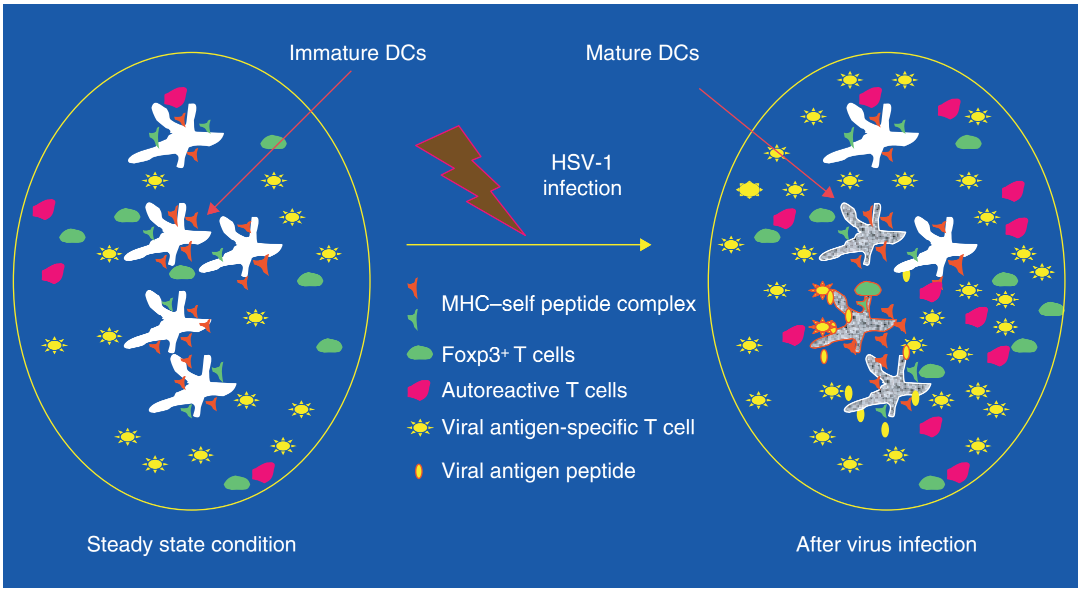

Figure 1. A possible model to explain how nTregs regulate HSV-1-specific T-cell response. In the steady state condition, nTreg are constantly activated as a result of their low-affinity T-cell receptor interactions with self-peptide-MHC complexes expressed on immature dendritic cells. These nTreg prevent the activation of autoreactive effectors (i.e., maintain peripheral tolerance). After virus infection, DCs mature (enhanced expression of costimulatory molecules) as a result of Toll-like receptor stimulation and present both viral-antigenic and self-antigenic peptides in association with MHC class I and II molecules, respectively. Thus, the same DCs involved in the induction of viral antigen-specific T-cell effector function will also enhance the potency of self-antigen reactive nTreg. These latter cells exert bystander immunosuppressive effects against the emerging virus-specific effectors hence reducing the magnitude of the response.
DC: Dendritic cell; HSV: Herpes simplex virus; nTreg: Naturally occurring CD4$^+$ Foxp3 pos T cell.

infections that suppress the immune system. Thus, a flare up of TB can occur in those infected with HIV [56], but it is not known if any change in Treg function accounts for this outcome.

8. Expert opinion

Overwhelming evidence shows that regulatory cells of various types play a key role in the outcome of infection by most and conceivably all pathogens. At a mechanistic level, it is quite well understood how Treg operate *in vitro*, but where and how they function in the host itself remains largely unknown. It is particularly difficult to explain is how pre-existing nTreg are signaled by invading pathogens to influence innate and adaptive immune events. Thus, the consensus opinion is that most nTreg are self-reactive. Moreover, even if some of them are pathogen specific or self cross-reactive, their frequency within the repertoire is likely to be very low. In addition, few Tregs of any specificity are expected to be already present at most entry sites of infection. Thus, with acute rapidly dividing infectious agents, antigen-specific recognition by nTreg at entry sites is unlikely to occur. Instead, with acute pathogens recruiting and activating self-reactive nTreg probably occurs and these act primarily in a polyclonal fashion in lymphoid tissues draining infection sites. This may occur by pathogens activating DC that express self-peptide–MHC II complexes, which in turn activate self-reactive nTreg (**Figure 1**). The activation could proceed either by the induction of inflammatory mediators or in some instances because the agents express innate immune activators, such as TLR ligands. Expression of TLR5 ligand by a pathogen results in nTreg activation [28], although TLR2 is reported to result in transient deactivation of nTreg [29]. In some instances, direct infection of nTreg may result in their activation [22]. Some ideas how acute infection with HSV might result in nTreg activation and influence immunity is expressed in **Figure 1**.

With agents that replicate less rapidly and less abundantly there is time for more regulatory events to unfold. The induction of cytokines such as IL-10 and TGFβ and in some instances the production by the pathogens of cytokine homologues, can result in the generation of at least two types of Treg that in this instance are antigen specific. Firstly, non-regulatory pathogen-specific T cells could be converted into antigen-specific nTreg and Foxp3^{neg} Tr1-type adaptive cells could also be generated. Both events have been demonstrated as possible *in vivo*, although perhaps not yet with infectious agents [11,16].

Conceivably, Treg induction or activation events that occur might differ between individuals and perhaps the circumstances of infection. Thus, a prompt reaction might serve to modulate the generation of effector function of protective CD8$^+$ and CD4$^+$ T cell responses to HCV, accounting for the subsequent chronic disease in such individuals. Unfortunately, longitudinal studies following

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events in individuals as they move from the acute to the chronic phase have not yet been done. With other infections, a prompt Treg response might prove valuable as it prevents immunopathology, the main cause of lesions [25]. This appears to be the state of affairs in SIV infection of disease-refractory AGMs [25]. At present, we are far from understanding the circumstances where Treg are favorable and in others where they are not. Such information is required if we are to exploit the use of Treg immunotherapeutically. We expect rapid progress in this field.

Another issue needing resolution is to define the sites at which nTreg exert their regulatory activity and how they subserve this function *in vivo*. For nTreg there is good evidence that they can act in lymphoid tissues and can modulate the generation and recall of adaptive immune responses [57]. We also believe that nTreg may limit the transport of protective components of immunity from lymphoid tissues where they are produced to peripheral sites of infection. However, whether or not nTregs that are readily detectable *in vivo* can function in inflammatory tissue remains an unresolved issue. Some maintain that inflammatory environments neutralize the function of nTreg, although how this might occur mechanistically remains unexplained. Moreover, studies that have isolated nTreg from inflammatory sites and shown their impaired function measured the latter *in vitro* rather than at the site itself.

Blunting of Treg by the inflammatory environment is perhaps more credible when regulatory cells act principally by making direct contact with the immune components they regulate. Thus, may be the usual case when nTreg act *in vivo* if *in vitro* studies are telling the correct story. However, if regulatory cells act by producing and exporting an abundance of anti-inflammatory cytokines, these may overwhelm any effect of pro-inflammatory cytokines. The production of abundant IL-10 is typical of Tr1 type regulators but this could also be occurring when natural or converted Treg are acting *in vivo*. It is also conceivable that some types of inflammatory events are more easily controllable by Tregs than are others. For example, T_H1 and T_H2 orchestrated reactions may be more amenable to control by certain types of Tregs than inflammatory events mediated by IL-17 producers. These issues require further study as they have an impact on future immunotherapy strategies for chronic infections. For example, we need to know what type of Treg to use if we are to successfully influence an established chronic infection. We would guess that Tr1-type cells, or perhaps nTreg converts, rather than conventional nTreg will be more useful.

The issue of how nTreg act is not only pertinent to control of infection. Thus, Tregs exporting certain cytokines at infection sites are likely to exhibit bystander suppressive events. These could be manifest in terms of increased susceptibility to other infections, and conceivably changed susceptibility to other inflammatory events such as allergies, autoimmunity and cancers. For example, excessive Treg effects to HCV in the liver have been associated with the development of hepatocellular carcinoma [58]. Bystander production of IL-10 and other regulatory products of Treg could result in suppression of allergic disease as may be happening in some parasitic disease [59]. Chronic infections in some instances might result in the redistribution of Treg to infection sites, thus reducing their presence at sites subject to autoinflammatory events. On the other hand, pathogen-activated Treg may result in the suppression of autoimmune disease as some have advocated to occur with Schistosoma infection [3].

We believe it is early days when it comes to understanding the relevance and especially exploiting the information about Treg and infection. Modulating Treg when they are harmful can result in more effective vaccines as observed in at least two instances [40,45]. We also expect that adoptive immunotherapy with Treg may in some chronic infections become a future approach and other means of changing Treg function. It will be important if this proves true to decide which type of Treg to use and to define ways to make the adoptive transfers to optimally seek the target tissue, be this the infection or lymphoid sites. Some recent approaches using the appropriate subsets of DC or modulators such as vitamin D to make T cells go to the skin appear to us to have great promise [60].

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RIGHTSLINK

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